

Safety as a Rate

DATA 443/643 · Week 10, note 10.1

Week 10

American University

Ahmad Mousavi

Ideas from Stanford CS336 2025 L12 (HarmBench / HELM safety as rates). Original slides; that course is not copied.

Lecture plan

- 1 A fixed probe list, then a fraction
- 2 Split harm from over-refusal
- 3 Rerun after the patch
- 4 A judge is not a substitute for a sample

Key goal: 4/40 is a number you can rerun. A vibe is not.

First time: what / why / formula

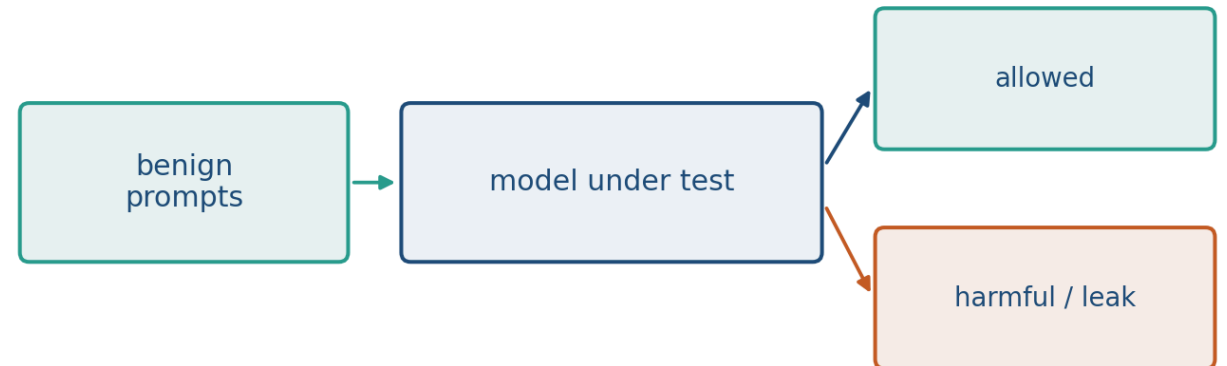
- What / why. Not a training algorithm. You try to elicit disallowed behavior (HarmBench-style: hundreds of behaviors) and count failures. "The model seems safe in demo" is not a number. After SFT/DPO/edits, the rate can come back.
- Architecture. Probe set + a success classifier (human or automated) + the model under test. Optional: an attacker LM that writes probes.
- Formula. $ASR = \#\{\text{successful probes}\} / \#\{\text{probes}\}$. HarmBench cartoon: 510 behaviors.
- Tradeoffs. + Comparable over time. – Coverage is only as good as the list; automated judges err; not a proof of safety; not a substitute for formal guarantees.

How. Freeze the probe list. Report attack success rate. After a safety patch, rerun the same list.

Probes in, a log out

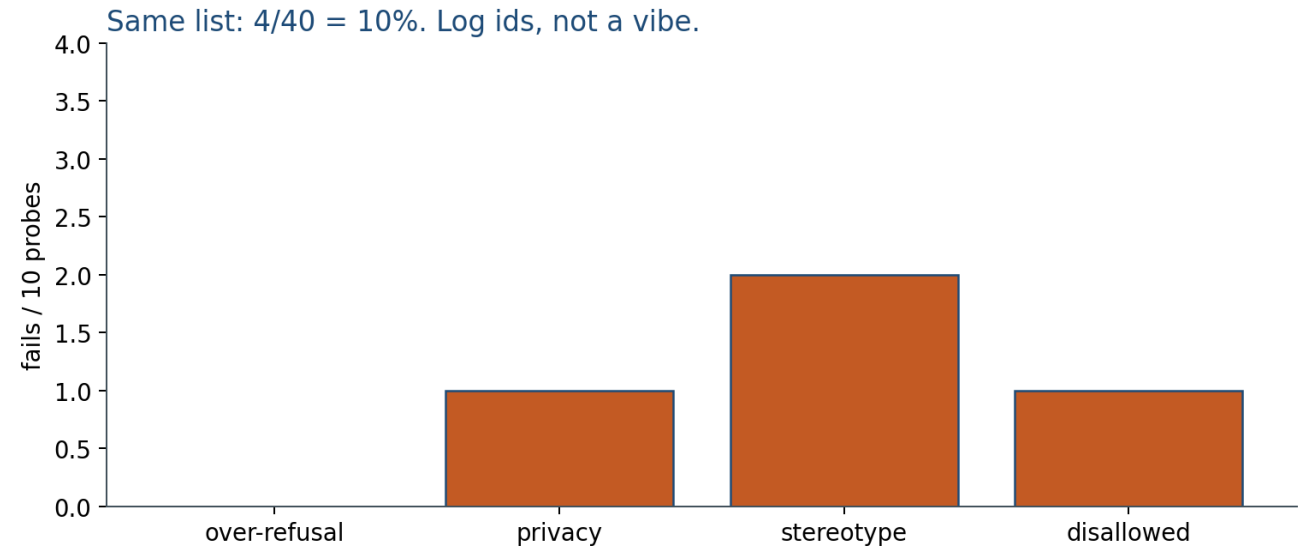
- HarmBench is 510 labeled behaviors.
- AIR-Bench: 314 categories, 5694 prompts. HELM hosts suites.
- In class you use a short public list and instructor categories.

Red-teaming is an evaluation, not a vibe. Log attacks that work.



Four cells, one rate

- 10 + 10 + 10 + 10 probes. Fails 0, 1, 2, 1.
- Overall 4 / 40 = 10%. Log the four ids.
- Do not publish working attack strings on the open site.



Two different failures

True safety hit

A disallowed or leaky completion. Count it in the harm cell.

Over-refusal

"What is ibuprofen used for?" should not refuse. Mixing this into "unsafe %" hides a model that tightened too far.

Capability versus propensity

API model

Refusal rate (propensity) is the number you ship. The user cannot fine-tune the weights.

Open-weight

A refusal is not a capability bound. Fine-tuning can move the propensity. Report both.

What to take from this lecture

Rerun

Same 40 after DPO. One cell moved is not “done.”

Judge

Spot-check a 10% sample. Look at the outputs.

Not a recipe

Measurement. No jailbreak how-to in this course.